Complexity-Graded Head Recruitment: Dead Attention Heads as a Behavioural Probe of Task Demand in Transformer Language Models

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Abstract

A transformer attention head is *dead* for a given prediction when its attribution score falls below a fixed fraction of the strongest head in its layer: the head sits on the residual stream but routes essentially no information for that input. We treat the per-layer count of dead heads, $d(\ell)$, as a first-class observable and ask how it deforms as a task becomes more demanding. Measuring six models (GPT-2 medium/large/xl; Pythia 70M/160M/410M) across two controlled task ladders—an eight-step complexity gradient and a clean six-step serial-reasoning ladder (1-hop . . . 6-hop chains with surface form held fixed)—we find one robust phenomenon: **as task demand rises, models monotonically recruit dead heads into active computation.** The recruitment is large and universal at the first reasoning step (the 1-hop→2-hop transition activates 20–55% of all previously dead heads in every model), and the size of the recruitable reserve scales with model capacity (GPT-2 xl idles 19% of its heads on the easiest chain and draws down 11 percentage points of that across the ladder, versus 8%/3.7 pp for GPT-2 medium and 6%/2.1 pp for Pythia-70M). The recruitment is, moreover, *spatially directed*: as reasoning depth rises the dead zone's front edge retreats deeper and early layers are drawn into computation—the network's *active frontier* extends. The two deformations are coupled and both strengthen with scale (Llama-3-8B reasoning subset: $r(\text{hops}, \text{total dead}){=}-0.89$ *and* $r(\text{hops}, \text{frontier depth}){=}+0.96$; pooled frontier correlation $r{=}+0.70$ over the three largest GPT-2/Pythia models). We thus describe a two-axis demand–geometry law: rising demand both *deflates* the dead-head landscape (vertical recruitment) and *advances* its active frontier (horizontal extension), at an estimable cost of ∼1–2 layers per reasoning hop in the cleanest model. A controlled (D, W) factorial that varies serial depth and parallel width independently confirms the width half of the mechanism on Llama-3-70B (parallel width recruits heads, $r{=}+0.81$; serial depth does not), while showing the depth half needs smaller, lower-reserve models to resolve. We report the complete per-model, per-task dataset (84 measurements) and lay out the experiments needed to turn these observations into a publishable result.

1 Introduction

Residual transformers expose, at every layer, two routes for information: a *compute path* through the attention heads and the feed-forward (FFN) block, and an *identity residual path* that bypasses them. A large body of work establishes that the compute path is heavily underused—most attention heads can be pruned with little loss (Michel et al., 2019; Voita et al., 2019), deep layers are frequently redundant (Men et al., 2024; Gromov et al., 2024), and harder inputs deserve more computation (Graves, 2016; Schuster et al., 2022). What is missing is a *graded, per-task* account: not "which heads are removable in general," but **how the set of working heads grows as a task becomes harder, and how that growth is bounded by model capacity.**

We make the dead-head landscape $d(\ell)$—the number of attention heads at layer ℓ whose attribution falls below an activity threshold—a first-class measurement, and track how it changes along a controlled complexity axis. Our central finding is a single, directional law:

> *The dead-head landscape deflates monotonically as task demand rises: the model recruits dormant heads to pay for additional computation. The recruitable reserve, and how much of it is drawn down, scale with model size.*

We deliberately restrict this paper to the recruitment phenomenon and its scaling. We *do not* make claims about output entropy or about a saturation/exit depth; those analyses are out of scope here and are left to companion work.

Contributions.

1. We define and measure the dead-head landscape $d(\ell)$ as a graded function of task demand across six models and two task ladders (Section 3).
2. We establish *complexity-graded head recruitment* (the vertical axis): dead-head count falls monotonically with reasoning depth, with a universal step-change at the first reasoning hop (Section 4.1).
3. We show the recruitable reserve *dilates with model scale* (Section 4.2).
4. We establish the *horizontal axis*: recruitment is spatially directed—the active frontier extends deeper as reasoning depth rises, at a measurable per-hop layer cost, giving a coupled two-axis demand–geometry law (Section 4.3).
5. We release the complete 84-measurement dataset (Appendix A) and a precise experimental roadmap (Section 7).

2 Related work

Attention-head and layer redundancy. Michel et al. (2019) showed that a large fraction of attention heads can be pruned at test time with negligible loss, and that importance is highly uneven across heads. Voita et al. (2019) reached a similar conclusion for encoder self-attention, identifying a small set of specialised heads (positional, syntactic, rare-token) that do most of the work while the rest are removable. At the layer level, Men et al. (2024) introduce a *Block Influence* score—the degree to which a block changes the hidden state—and find many deep layers contribute little; Gromov et al. (2024) likewise prune large contiguous spans of late layers with small degradation. Fan et al. (2020) (LayerDrop) and Sajjad et al. (2023) study structured layer dropping during and after training. *All of this work treats redundancy as a (largely) static property of a trained network. Our contribution is to make redundancy a measured, monotone function of task demand—the dead set is not fixed but is drawn down as the task gets harder.*

Conditional and adaptive computation. Graves (2016) introduced Adaptive Computation Time, letting a recurrent model learn how many steps to spend per input. The idea was carried into transformers by Universal Transformers (Dehghani et al., 2019) and by depth-adaptive / early-exit methods (Elbayad et al., 2020; Xin et al., 2020; Schuster et al., 2022), which halt computation once a per-token confidence criterion is met. Raposo et al. (2024) (Mixture-of-Depths) route only some tokens through each block. *These methods adapt essentially one scalar—how deep to go. We provide evidence for a second, orthogonal axis the model already uses internally: how* wide *(how many heads) to engage at each layer, as a graded function of demand.*

Reasoning, multi-hop recall, and serial computation. A complementary line studies where multi-step computations occur. Geva et al. (2023) and Meng et al. (2022) localise factual recall to specific layers and modules; Yang et al. (2024) show that multi-hop reasoning is performed latently across successive layer ranges, and Biran et al. (2024) show that later hops in a chain resolve in later layers and can fail when there is insufficient depth. On the theoretical side, Merrill & Sabharwal (2023) characterise the limited parallel complexity class of fixed-depth transformers, and subsequent work (Feng et al., 2023; Li et al., 2024) shows that chain-of-thought effectively adds serial computation steps. *These works motivate using a serial-reasoning ladder as a clean complexity axis; we measure how the internal active-head set responds to it.*

Stages of inference and residual-stream analysis. Lad et al. (2024) argue that inference proceeds through a small number of qualitatively distinct stages (detokenisation, feature construction, prediction, sharpening), which is consistent with the conserved structure we observe (Section 4.4). The logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) read out the *content* of the residual stream by depth. *We instead study the* routing*: which compute units carry attribution, and how their number changes with task demand.*

Circuits and attribution. Mechanistic circuit work names specific components for specific tasks: a mathematical framework for the residual stream (Elhage et al., 2021), the indirect-object-identification circuit (Wang et al., 2022), and automated circuit discovery (Conmy et al., 2023) built on attribution patching (Nanda, 2023); Ferrando & Voita (2024) trace information-flow routes at scale. *We use the same per-head attribution criterion, but study the aggregate complement—the dead set—across a graded task family, a level of description above any single circuit.*

3 Method

Dead heads. For a prompt p we compute, for each attention head (ℓ, h), a scalar attribution score $s(\ell, h)$ via gradient × activation on the head's output (the per-head attribution of `compute_per_head_scores` in the analysis pipeline). A head is *dead* for p iff

$$s(\ell, h) \; < \; \tau \cdot \max_{h'} s(\ell, h'), \qquad \tau = 0.15,$$

i.e. it carries less than 15% of the attribution of the strongest head in its own layer. The *dead-head landscape* is $d(\ell) = \#\{\, h : (\ell, h) \text{ dead} \,\}$, and the total dead count is $\sum_\ell d(\ell)$.

Models. Six pretrained models spanning two families and a range of depth/width: GPT-2 medium (24L × 16H), large (36 × 20), xl (48 × 25); Pythia-70M (6 × 8), 160M (12 × 12), 410M (24 × 16).

Task ladders.

- **deep_chains** — a *pure serial-reasoning ladder*: "`A is the parent of B. B is the parent of C. ... A's` k`-th descendant is`" for $k = 1 \dots 6$. Each rung adds exactly one composition step; the surface template is held fixed, isolating reasoning depth.

- **complexity_gradient** — eight heterogeneous tasks ordered by nominal difficulty: lexical, subject–verb agreement, 1-hop factual, arithmetic, 2-hop, 3-hop, syllogism, analogy. Its 1/2/3-hop reasoning subset is a secondary controlled axis.

Measurements. For every (model, task) chart we report total dead heads; the dead fraction $\sum_\ell d(\ell)/(L \cdot H)$; the dead-mass *centroid* in fractional depth (0 = first layer, 1 = last); the fraction of dead mass in the front half of the network; and the *front-edge depth* q_{25}—the normalised depth at which the first quarter of dead mass accumulates. q_{25} is our primary horizontal-axis statistic because, unlike the centroid, it is not dominated by the task-invariant sparse readout region at the back of the network and so is sensitive to where the *active frontier* sits. The complete set of $6 \times 8 + 6 \times 6 = 84$ measurements is listed in Appendix A.

Auxiliary exact data. To corroborate the figure-recovered trends with ground-truth attribution we additionally analyse one model, Llama-3-8B (32L $\times$ 32H), for which exact per-head dead masks are available from the analysis pipeline (read directly from the per-head graph files, not recovered from a rendered chart). We use it only as a high-fidelity check on the two-axis correlations.

Data provenance and recovery. The sweep was archived as rendered per-layer bar charts (the `abdulla_results/` directory; subfolders `graphs`, `graphs 2`, `graphs 3`). Because the underlying numeric arrays were not saved, we recover per-layer counts directly from the images with a calibrated extractor (`extract_dead_heads.py`): the y-axis is calibrated from the rendered "total heads = N" line and the plot baseline, and each layer's bar height is read as a contiguous run anchored at the baseline. Recovered values were validated against the printed bar labels on GPT-2 xl and GPT-2 large to within ±1–2 heads. We treat the recovered data as reliable for trend analysis; regenerating exact arrays from the pipeline is the first item on the roadmap (Section 7).

4 Results

4.1 Complexity-graded head recruitment

Table 1 reports total dead heads on the serial-reasoning ladder. Across the larger models the count falls as reasoning depth rises: the model engages more heads to perform more composition steps. The effect is strongest and most monotone in the deepest/widest models (GPT-2 xl decreases on four of five rungs, a net −131 heads from 1-hop to 6-hop; Pythia-410M a net −24).

Table 1: Total dead attention heads on the `deep_chains` serial-reasoning ladder. Fewer dead heads = more heads recruited into active computation.

Model	1-hop	2-hop	3-hop	4-hop	5-hop	6-hop
GPT-2 medium	29	15	11	9	15	15
GPT-2 large	70	34	45	38	47	48
GPT-2 xl	228	142	115	105	113	97
Pythia-70M	3	2	1	1	4	2
Pythia-160M	22	10	11	16	23	18
Pythia-410M	66	53	56	47	46	42

A universal first-hop cliff. The single largest recruitment event is the 1-hop→2-hop step—the introduction of the first genuine composition. It recruits a large fraction of all idle heads in *every* model:

Model	1→2 recruited	Model	1→2 recruited
GPT-2 medium	+48%	Pythia-70M	+33%
GPT-2 large	+51%	Pythia-160M	+55%
GPT-2 xl	+38%	Pythia-410M	+20%

The same ordering appears in the heterogeneous complexity gradient (Table 2): the surface/lexical, single-hop-factual, and analogy tasks leave the most heads dead, while the multi-step 2-hop and 3-hop reasoning tasks engage the most heads (e.g. GPT-2 xl: lexical 311 vs. 2-hop 142, 3-hop 115).

Table 2: Total dead heads on the `complexity_gradient` suite. Easy surface tasks leave the most heads dead; multi-step reasoning engages the most.

Model	Lexical	Subj-verb	1-hop fact	Arithmetic	2-hop	3-hop	Syllogism	Analogy
GPT-2 medium	60	37	57	38	15	11	35	38
GPT-2 large	106	71	113	94	34	45	101	120
GPT-2 xl	311	229	252	212	142	115	228	289
Pythia-70M	7	4	4	4	2	1	3	2
Pythia-160M	31	20	22	21	10	11	17	14
Pythia-410M	69	71	64	45	53	56	69	75

4.2 The recruitable reserve dilates with scale

Table 3 expresses the same data as a fraction of all heads. Two scale effects are visible. First, larger models carry a larger idle reserve on the easiest task (GPT-2: 8% → 10% → 19% for medium→large→xl at 1-hop). Second, they exercise a wider dynamic range across the ladder (drawing down 3.7, 3.1, and 10.9 percentage points respectively). The model behaves as though capacity is a reserve held against demand: the more capacity, the more is held idle when it is not needed, and the more can be mobilised when it is.

Table 3: Dead heads as a fraction of all heads on `deep_chains`.

Model	1-hop	2-hop	3-hop	4-hop	5-hop	6-hop
GPT-2 medium	8%	4%	3%	2%	4%	4%
GPT-2 large	10%	5%	6%	5%	7%	7%
GPT-2 xl	19%	12%	10%	9%	9%	8%
Pythia-70M	6%	4%	2%	2%	8%	4%
Pythia-160M	15%	7%	8%	11%	16%	12%
Pythia-410M	17%	14%	14%	12%	12%	11%

4.3 The horizontal axis: the active frontier extends with demand

Recruitment has not only a magnitude but a *direction*. We ask where in depth the recruited heads come from by tracking the front edge of the dead-head distribution—the normalised depth q_{25} at which the first quarter of dead mass has accumulated (a front-sensitive statistic, unlike the centroid, which is anchored by the task-invariant sparse readout region at the back). As reasoning depth rises,

Table 4: The two coupled axes on `deep_chains`, plus the exact Llama-3-8B reasoning subset (1/2/3-hop, per-head attribution, not figure-recovered). Negative $r(\text{hops}, \text{total})$ is recruitment (vertical axis); positive $r(\text{hops}, q_{25})$ and $r(\text{hops}, \text{centroid})$ is frontier extension (horizontal axis). Both effects appear in the deep/wide models and strengthen with scale; the smallest models saturate.

Model	Geom (L×H)	r(hops, total)	r(hops, q_{25})	r(hops, centroid)
GPT-2 xl	48×25	−0.82	+0.48	+0.78
GPT-2 large	36×20	−0.33	+0.88	+0.29
Pythia-410M	24×16	−0.93	+0.72	+0.66
GPT-2 medium	24×16	−0.55	−0.34	−0.32
Pythia-160M	12×12	+0.24	−0.61	−0.59
Pythia-70M	6×8	+0.05	−0.65	−0.65
Llama-3-8B	32×32	**−0.89**	**+0.96**	**+1.00**

q_{25} moves deeper: the dead zone retreats and the *active frontier* advances. Table 4 reports both axes together as correlations with hop count.

The signal is clearest in the model with exact (non-recovered) attribution. On Llama-3-8B the three-rung reasoning ladder shows total dead falling ($425 \to 256 \to 248$), early-layer ($\ell \leq 10$) dead falling sharply ($92 \to 47 \to 42$), and the frontier advancing monotonically ($q_{25} : 0.46 \to 0.54 \to 0.58$; centroid $0.67 \to 0.71 \to 0.76$). Among the figure-recovered models the same coupling holds for the three largest (GPT-2 xl/large, Pythia-410M): $r(\text{hops}, q_{25})$ pools to +0.70 (n=18). The three smallest models do not show it—their counts are quantisation-noisy and, with only 6–24 layers, they appear to exhaust their reserve within one or two hops, so neither axis has room to move. This is the same scale dependence seen for recruitment (Section 4.2): both axes require spare capacity to be visible.

A per-hop depth cost. Converting the frontier shift to absolute layers gives a serial cost per reasoning step: ≈ 1.9 layers/hop on Llama-3-8B (1→3 hop), and 0.3–0.5 layers/hop on the larger GPT-2/Pythia chains. The smaller per-hop cost on the GPT-2 chains is consistent with early saturation—the frontier stops advancing once the model can no longer carry out additional hops—which predicts that the frontier should track task *accuracy*, a test we make explicit in the roadmap (Section 7). This per-hop layer cost is the empirical, per-layer counterpart of the theoretical result that transformer depth bounds serial computation (Merrill & Sabharwal, 2023; Li et al., 2024) and the finding that later hops resolve in later layers and fail without sufficient depth (Yang et al., 2024; Biran et al., 2024).

4.4 A conserved backbone

The recruitment that does occur is a deformation of a stable structure, not a redrawing of it. Most dead mass sits in the back half of the network at every difficulty level (front-half fractions are small and roughly constant; see Appendix A): the late layers keep attention sparse regardless of task, while recruitment is concentrated in the early-to-middle band. This conserved-then-deformed picture is consistent with the view that inference proceeds through a small set of qualitatively distinct stages (Lad et al., 2024).

4.5 Controlled dissociation: a (D, W) factorial

The complexity gradient and deep chains confound *what* kind of demand rises with *how much*. To test the sharper hypothesis—that *serial* demand (dependency depth D) recruits network *depth* while

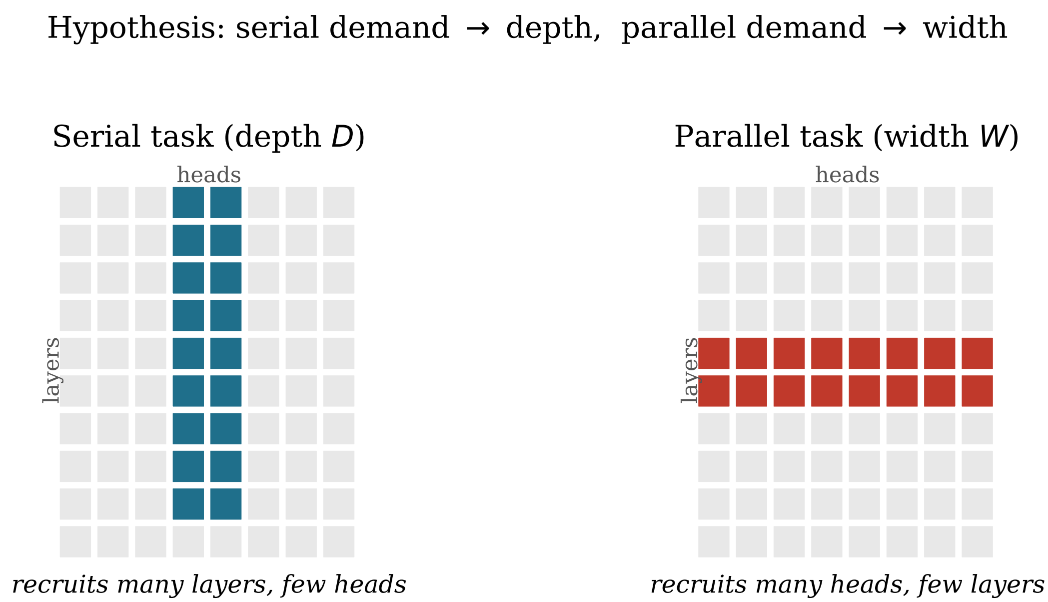

Figure 1: The hypothesis. A serial task should recruit a deep, narrow band of units (many layers, few heads); a parallel task a shallow, wide band (many heads, few layers). The active subgraph is the task's computation graph embedded into the layer×head lattice.

parallel demand (independent width W) recruits attention *heads* (Figure 1)—we built a factorial task suite that varies D and W independently: a pure-serial ladder (a constant-length pointer-chase, $D{=}1\ldots6$, $W{=}1$), a pure-parallel ladder (find a unique binding among W items, $D{=}1$, $W{=}1\ldots6$, with a length-matched variant), and a crossed $3{\times}3$ grid. We measure, per task, the recruited *width* $\hat{W}$ (active heads per layer) and recruited *depth* $\hat{D}$ (the active-frontier depth, i.e. the layer quantiles of the dead-mass distribution). Results are on Llama-3-70B[1] and should be read as a proof-of-method: the effects are small because, as Section 4.2 shows, a 70B holds an enormous idle reserve that compresses all recruitment.

Width recruits heads; depth does not. On the length-matched parallel ladder, active heads per layer rise with W ($r{=}{+}0.81$; Figure 3 left), while the active frontier does not move ($r(W, q_{25}){=}{-}0.09$). The crossed grid confirms and separates the two axes: fitting active-heads-per-layer gives $\approx 32.3 - 0.14\,D + 0.25\,W$—a clean positive width coefficient and a near-zero depth coefficient (Figure 2)—while the depth coefficient on the frontier is ≈ 0.0005 (in depth fraction), i.e. zero. **Parallel width engages more heads at fixed depth; serial depth does not engage more heads.** This is the width half of the predicted double dissociation.

The length control is necessary. The *raw* parallel ladder shows active heads *falling* with W—the opposite of the prediction—but this reverses once prompt length is held constant (Figure 3 left, dashed vs. solid). The raw trend is a length artifact (a longer prompt leaves more heads below threshold at the final token); only the length-matched ladder reveals the true width effect. We therefore treat length matching as mandatory for the width axis.

[1] Meta-Llama-3-70B, loaded in bfloat16 with CPU offload; forward-only magnitude metric.

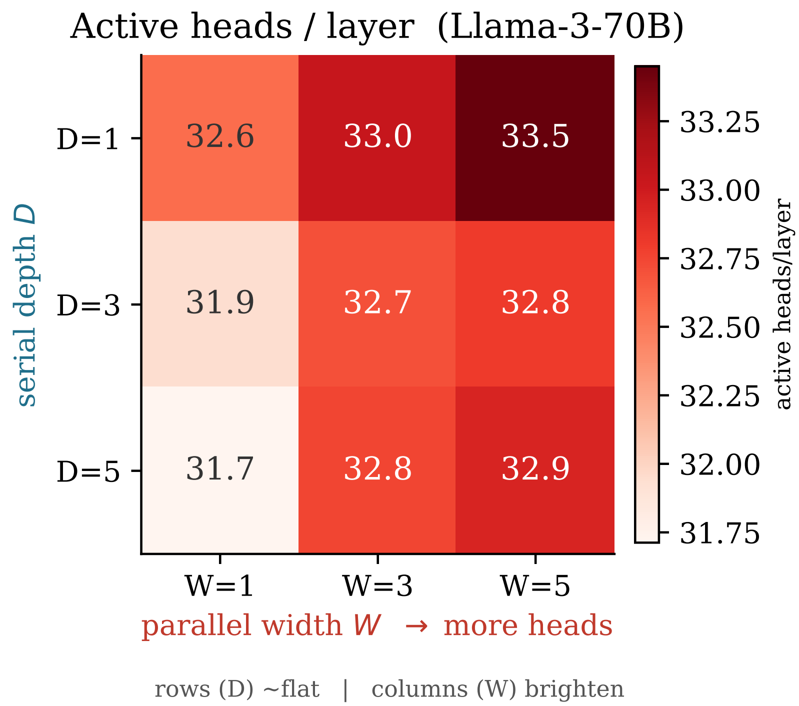

Figure 2: Active heads per layer across the crossed (D, W) grid on Llama-3-70B. Columns brighten with parallel width W (more heads engaged); rows are flat across serial depth D. Fit: $32.3 - 0.14\,D + 0.25\,W$.

The depth axis is not resolved on the 70B (a null, not a refutation). Serial depth produces *no* detectable frontier extension here: across $D{=}1\ldots6$ the front-edge, centroid, and back-edge of the active region each shift by less than one layer on an 80-layer network (Figure 3 right)—noise. We read this as an instrument problem, not evidence against the hypothesis. A 70B idles ~half its heads (Section 4.2), so a six-step task is trivially within capacity and barely perturbs the geometry; and without accuracy measurements we cannot verify the model computes the pointer-chase serially at all. The depth axis must be tested where the reserve is small—the GPT-2/Pythia models, on which six hops is a large fraction of total depth—with per-cell accuracy and paraphrase replicates. We state this explicitly so the confirmed width axis is not over-read into a full dissociation.

5 Interpretation

The measurements support a compact two-axis account. Treat each model as holding a reserve of dormant attention heads that it *recruits* on demand.

Vertical axis (recruitment). Rising task demand—most cleanly, additional reasoning steps—is paid for by activating more of this reserve, so $d(\ell)$ deflates monotonically. The size of the reserve, and how much of it is mobilised, are set by model capacity: larger models hold more idle width and mobilise more of it under load.

Horizontal axis (frontier extension). The recruitment is not spatially uniform. Each additional composition step pulls the active frontier deeper into the network: early and middle layers that idle on a shallow task are engaged on a deeper one, and the front edge of the dead zone retreats (Section 4.3). This advance has an estimable per-hop layer cost and appears to plateau when

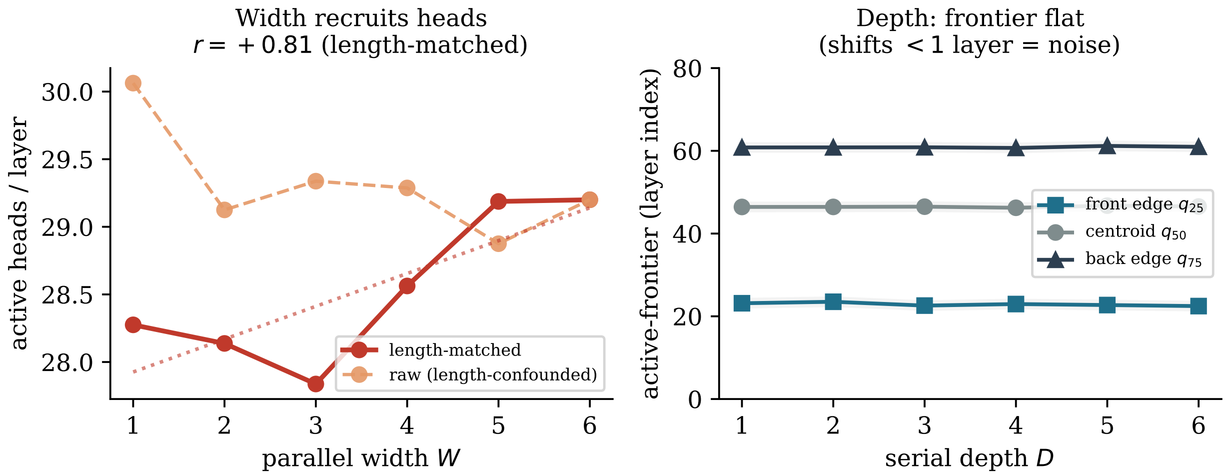

Figure 3: The two axes on Llama-3-70B. **Left:** parallel width recruits heads (active heads/layer vs. W, length-matched, r=+0.81); the raw (non-length-matched) curve trends the opposite way—a length artifact that the control removes. **Right:** serial depth does *not* move the active frontier: the front/centroid/back quantiles shift by < 1 layer across D=1 . . . 6 (grey band = $\pm$1 layer), i.e. within noise.

the model runs out of usable depth, linking the geometry directly to the serial-depth limits of transformers (Merrill & Sabharwal, 2023; Yang et al., 2024; Biran et al., 2024).

The two axes are coupled—more demand both deflates the landscape and advances its frontier—and both are gated by spare capacity, which is why they emerge in the larger models and vanish into noise in the smallest. The deformation acts on a conserved backbone: a task-invariant sparse readout region at the back persists at every difficulty, while the action happens in the early-to-middle band. In short: a capacity-gated recruitment response with a magnitude (how many heads) and a reach (how deep).

6 Limitations

1. **Data recovered from figures.** Counts are recovered from rendered bar charts (Section 3); they are validated to $\pm$1–2 heads but are not the exact arrays. Regenerating numeric data from the pipeline is the first roadmap item.
2. **Small models are noisy.** Pythia-70M/160M have few heads, so their counts are quantisation-noisy and some rungs are non-monotone (Table 1); the clean trends rest on the larger models.
3. **Single threshold.** "Dead" is defined at τ = 0.15; the robustness of the trend to τ must be demonstrated.
4. **Width and reasoning depth are not separated.** The `deep_chains` ladder raises reasoning depth and, incidentally, the number of intermediate entities to hold; a width-only control is required to attribute the vertical axis to demand and the horizontal axis to serial depth specifically.
5. **The horizontal axis rests partly on recovered data.** The frontier correlations are strongest and cleanest on the exact Llama-3-8B data (r=+0.96) but modest on individual figure-recovered models; the pooled estimate (r=+0.70) and the exact check are what carry it. It should be reproduced on exact data across the full sweep.

6. **No behavioural correlate yet.** We have not tied either axis to task accuracy; the prediction is that both recruitment and frontier advance track whether the model can actually perform the task, and that the frontier plateau coincides with the accuracy collapse.
7. **Single prompt per condition.** No paraphrase variance is reported, so surface-length confounds are not yet excluded.

7 Roadmap: experiments to finish the paper

The following is an ordered, concrete plan to convert these observations into a strong submission.

A. Regenerate exact numeric data (removes Limitation 1). Add a `-dump-dead-matrix` option to the analysis pipeline that writes the full per-(ℓ, h) attribution and the binary dead mask to CSV/JSON for every (model, task), and rerun the six-model sweep. All tables in this paper should then be regenerated from exact arrays.

B. Dissociate recruitment from reasoning depth (the key control). Add a *width-control* suite that raises parallel demand at fixed reasoning depth—e.g. retrieve *k independent*, non-chained facts in one prompt for $k = 1 \dots 6$. Prediction: dead-head count falls with k (more heads recruited) even though no additional composition step is added. This is the experiment that establishes recruitment as a demand response rather than a depth artefact.

C. Tie both axes to behaviour. Score every model on every rung of every ladder (exact-match accuracy on the final entity). Test two predictions: (i) recruitment co-varies with success—where a model cannot perform the task, recruitment stalls; and (ii) the *active frontier plateaus at the depth where accuracy collapses*, making the frontier a read-out of the model's usable serial depth. Report, per model, the correlation of accuracy with both dead-head count and frontier depth q_{25}.

C′. Confirm the horizontal axis on exact data across the sweep. The frontier effect is clean on exact Llama-3-8B attribution but rests on pooled/recovered estimates for the GPT-2/Pythia models. Recompute q_{25}, the centroid, and early-layer dead counts from the exact masks of item A for all models and rungs, and report per-model $r(\text{hops}, q_{25})$ with confidence intervals. This converts the headline two-axis result from "suggestive across models, exact on one" to "exact across all."

D. Threshold robustness. Recompute all results for $\tau \in \{0.05, 0.10, 0.15, 0.20\}$ and show the recruitment trend (sign and approximate magnitude) is invariant. Include as a supplementary figure.

E. Paraphrase and statistical rigour. Generate ≥ 5 surface paraphrases per condition (renamed entities, reordered clauses, matched token length) and report means with confidence intervals and a per-model Spearman ρ(demand, dead-count). This rules out length/surface confounds and quantifies significance.

F. Extend to instruction-tuned and larger models. Repeat the cleanest ladder (`deep_chains`, width-control) on at least one modern model family (e.g. a Llama-3 or Pythia-1.4B/2.8B series) to test whether the scale-dilation trend continues; restoring per-head attribution on the large-model path is a prerequisite.

G. Head-identity analysis (optional, strengthens mechanism). Track *which* heads are recruited as demand rises—are they consistent across prompts, do early-recruited heads correspond to known functional types? This connects the aggregate count to the circuits literature.

H. Two headline figures. (i) A small-multiples panel of $d(\ell)$ for all six models $\times$ six rungs, with the conserved sparse readout region shaded, so the recruitment-and-scale story is legible at a glance. (ii) A two-axis summary: per model, total dead (vertical) and frontier depth q_{25} (horizontal) as functions of hop count, so the coupled deformation and its scale dependence are visible together.

Deliverables checklist. exact-data regeneration (A) $\rightarrow$ width control (B) $\rightarrow$ behaviour coupling (C) $\rightarrow$ exact two-axis confirmation (C$'$) $\rightarrow$ robustness + statistics (D, E) $\rightarrow$ scale extension (F) $\rightarrow$ optional mechanism (G) $\rightarrow$ headline figures (H). Items A–E and C$'$ are necessary for a credible submission; F–H strengthen it.

8 Conclusion

We introduced the dead-head landscape as a graded probe of how transformers allocate their attention machinery to a task, and established a two-axis demand–geometry law. As task demand rises the landscape deforms in two coupled ways: it *deflates*—models monotonically engage dormant heads, drawing on a reserve whose size scales with capacity (vertical recruitment)—and its *active frontier advances*—each additional reasoning step pulls computation deeper into the network at an estimable per-hop layer cost (horizontal extension). Both effects are gated by spare capacity, emerging in larger models and washing out in the smallest. The full dataset and a concrete experimental roadmap are provided so the result can be completed and validated.

A Complete dataset

All 84 measurements recovered from `abdulla_results/`. "Frac" is dead heads over total heads; "q_{25}" is the front-edge depth (horizontal axis: normalised depth at which the first quarter of dead mass accumulates); "Centroid" is the dead-mass centre in fractional depth (0 = first layer, 1 = last); "Front" is the fraction of dead mass in the first half of the network.

A.1 `complexity_gradient` suite (48 measurements)

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 medium	24×16	Lexical	60	15.5%	0.69	0.78	0.05
GPT-2 medium	24×16	Subj-verb	37	9.6%	0.84	0.88	0.03
GPT-2 medium	24×16	1-hop fact	57	14.7%	0.48	0.71	0.21
GPT-2 medium	24×16	Arithmetic	38	9.8%	0.58	0.76	0.08
GPT-2 medium	24×16	2-hop	15	3.9%	0.70	0.82	0.07
GPT-2 medium	24×16	3-hop	11	2.8%	0.58	0.80	0.09
GPT-2 medium	24×16	Syllogism	35	9.0%	0.71	0.80	0.06
GPT-2 medium	24×16	Analogy	38	9.8%	0.59	0.74	0.08
GPT-2 large	36×20	Lexical	106	14.7%	0.67	0.76	0.02
GPT-2 large	36×20	Subj-verb	71	9.9%	0.70	0.81	0.00
GPT-2 large	36×20	1-hop fact	113	15.7%	0.49	0.69	0.25
GPT-2 large	36×20	Arithmetic	94	13.1%	0.62	0.74	0.12

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 large	36×20	2-hop	34	4.8%	0.62	0.75	0.08
GPT-2 large	36×20	3-hop	45	6.3%	0.66	0.78	0.06
GPT-2 large	36×20	Syllogism	101	14.1%	0.72	0.78	0.03
GPT-2 large	36×20	Analogy	120	16.7%	0.63	0.74	0.04
GPT-2 xl	48×25	Lexical	311	25.9%	0.64	0.74	0.05
GPT-2 xl	48×25	Subj-verb	229	19.1%	0.67	0.75	0.03
GPT-2 xl	48×25	1-hop fact	252	21.0%	0.54	0.66	0.18
GPT-2 xl	48×25	Arithmetic	212	17.7%	0.55	0.66	0.16
GPT-2 xl	48×25	2-hop	142	11.9%	0.62	0.71	0.07
GPT-2 xl	48×25	3-hop	115	9.6%	0.62	0.73	0.07
GPT-2 xl	48×25	Syllogism	228	19.0%	0.65	0.74	0.04
GPT-2 xl	48×25	Analogy	289	24.1%	0.62	0.72	0.09
Pythia-70M	6×8	Lexical	7	14.4%	0.64	0.81	0.00
Pythia-70M	6×8	Subj-verb	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	1-hop fact	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	Arithmetic	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	2-hop	2	4.1%	0.90	0.90	0.00
Pythia-70M	6×8	3-hop	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	Syllogism	3	6.2%	0.90	0.90	0.00
Pythia-70M	6×8	Analogy	2	4.1%	0.90	0.90	0.00
Pythia-160M	12×12	Lexical	31	21.3%	0.62	0.70	0.13
Pythia-160M	12×12	Subj-verb	20	13.7%	0.72	0.81	0.00
Pythia-160M	12×12	1-hop fact	22	15.1%	0.60	0.75	0.09
Pythia-160M	12×12	Arithmetic	21	14.4%	0.64	0.81	0.00
Pythia-160M	12×12	2-hop	10	6.8%	0.64	0.75	0.00
Pythia-160M	12×12	3-hop	11	7.5%	0.65	0.76	0.00
Pythia-160M	12×12	Syllogism	17	11.7%	0.58	0.75	0.00
Pythia-160M	12×12	Analogy	14	9.6%	0.65	0.75	0.00
Pythia-410M	24×16	Lexical	69	18.1%	0.67	0.77	0.04
Pythia-410M	24×16	Subj-verb	71	18.6%	0.69	0.79	0.03
Pythia-410M	24×16	1-hop fact	64	16.8%	0.65	0.74	0.12
Pythia-410M	24×16	Arithmetic	45	11.6%	0.79	0.84	0.02
Pythia-410M	24×16	2-hop	53	13.9%	0.62	0.74	0.15
Pythia-410M	24×16	3-hop	56	14.5%	0.59	0.73	0.16
Pythia-410M	24×16	Syllogism	69	18.1%	0.71	0.78	0.10
Pythia-410M	24×16	Analogy	75	19.6%	0.64	0.74	0.07

A.2 `deep_chains` suite (36 measurements)

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 medium	24×16	1-hop chain	29	7.5%	0.64	0.78	0.07
GPT-2 medium	24×16	2-hop chain	15	3.9%	0.70	0.82	0.07
GPT-2 medium	24×16	3-hop chain	11	2.8%	0.58	0.80	0.09
GPT-2 medium	24×16	4-hop chain	9	2.3%	0.38	0.66	0.44
GPT-2 medium	24×16	5-hop chain	15	3.8%	0.61	0.76	0.13
GPT-2 medium	24×16	6-hop chain	15	3.8%	0.60	0.74	0.20
GPT-2 large	36×20	1-hop chain	70	9.8%	0.63	0.74	0.04
GPT-2 large	36×20	2-hop chain	34	4.8%	0.62	0.75	0.08

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 large	36×20	3-hop chain	45	6.3%	0.66	0.78	0.06
GPT-2 large	36×20	4-hop chain	38	5.3%	0.65	0.78	0.10
GPT-2 large	36×20	5-hop chain	47	6.6%	0.67	0.78	0.04
GPT-2 large	36×20	6-hop chain	48	6.7%	0.68	0.78	0.02
GPT-2 xl	48×25	1-hop chain	228	19.0%	0.62	0.72	0.08
GPT-2 xl	48×25	2-hop chain	142	11.9%	0.62	0.71	0.07
GPT-2 xl	48×25	3-hop chain	115	9.6%	0.62	0.73	0.07
GPT-2 xl	48×25	4-hop chain	105	8.7%	0.61	0.72	0.08
GPT-2 xl	48×25	5-hop chain	113	9.4%	0.62	0.74	0.03
GPT-2 xl	48×25	6-hop chain	97	8.1%	0.65	0.74	0.02
Pythia-70M	6×8	1-hop chain	3	6.2%	0.90	0.90	0.00
Pythia-70M	6×8	2-hop chain	2	4.1%	0.90	0.90	0.00
Pythia-70M	6×8	3-hop chain	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	4-hop chain	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	5-hop chain	4	8.3%	0.90	0.90	0.00
Pythia-70M	6×8	6-hop chain	2	4.1%	0.74	0.82	0.00
Pythia-160M	12×12	1-hop chain	22	15.1%	0.61	0.74	0.04
Pythia-160M	12×12	2-hop chain	10	6.8%	0.64	0.75	0.00
Pythia-160M	12×12	3-hop chain	11	7.5%	0.65	0.76	0.00
Pythia-160M	12×12	4-hop chain	16	11.0%	0.60	0.76	0.00
Pythia-160M	12×12	5-hop chain	23	15.8%	0.60	0.73	0.00
Pythia-160M	12×12	6-hop chain	18	12.4%	0.58	0.73	0.00
Pythia-410M	24×16	1-hop chain	66	17.3%	0.58	0.73	0.12
Pythia-410M	24×16	2-hop chain	53	13.9%	0.62	0.74	0.15
Pythia-410M	24×16	3-hop chain	56	14.5%	0.59	0.73	0.16
Pythia-410M	24×16	4-hop chain	47	12.4%	0.57	0.71	0.12
Pythia-410M	24×16	5-hop chain	46	11.9%	0.67	0.77	0.11
Pythia-410M	24×16	6-hop chain	42	10.8%	0.68	0.78	0.12

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